MAKSIM SILCHENKO

mthylinao@gmail.com | Singapore | +44 77 2925 4349 | github.com/thylinao1 | linkedin.com/in/maksim-silchenko
Portfolio Page: https://thylinao1.github.io

EXPERIENCE

Orbuc Research, London, UK [Catalyst Agent Prototype] Feb 2026 - Apr 2026
Quantitative Research Intern

- Developed a time-series anomaly-detection model that classifies latent states with Hidden Markov Models to flag unstable, low-predictability conditions, validated across 5 out-of-sample windows where it held up in 4 of 5.
- Built a production-hardened Python pipeline that uses an LLM to classify events from live news streams, engineered for reliability with batched API calls, rate-limit handling, SQL persistence, and an automated test suite.
- Engineered the statistical analysis behind the firm's March 2026 research report, characterising 9 historical anomaly periods and identifying 4 statistically significant shifts in correlation structure.

The TCM Group, London, UK [Website + LLM Prototype] [GitHub] Jan 2026 - Apr 2026
Data & Analytics Consultant Intern (Bayes Business School Capstone) · Top of the Cohort

- Designed a causal-inference framework (RCT-style wait-list control, paired hypothesis tests) to measure leadership-training outcomes, presented it to TCM's senior leadership, and saw it adopted as the standard for ongoing programme evaluation.
- Built a cloud LLM scoring assistant (Vercel, OpenAI) that grades open-ended training responses 0 to 10 against rubrics, deployed with Zod schema validation and automatic retry so malformed model output is never returned.
- Built an automated evaluation harness that benchmarks the scorer against a hand-labelled gold dataset (0.90 Spearman correlation with human raters), then halved its scoring error with a cross-validated calibration that corrects the model's systematic bias.

Oxford Comma Advisory, London, UK Oct 2025 - Jan 2026
Data Analyst Intern

- Designed a two-tower recommendation system using learned embeddings for university matching, reaching 0.82 precision@5 and cutting consultant research time from 2 hours to 3 minutes.
- Built a feature-engineering pipeline feeding an XGBoost churn model and a Cox Proportional Hazards survival analysis on 800+ inquiries, reaching AUC 0.79 and identifying lead sources that convert 60% faster.
- Designed and analysed A/B tests on personalised outreach, lifting booking conversion from 18% to 27% (p = 0.004).

AWARDS & ACHIEVEMENTS

- **Russian National Mathematics Olympiad (MIPT): Winner**, top 0.015% nationally.
- **Jane Street 'Puzzle' Competition: Winner** (Feb 2026). **Bloomberg Trading Challenge: Captain**, +32% return vs Bloomberg WLS index.
- **Top of the Cohort**: 'Quantitative Methods & Analytics', 'AI & Big Data', 'Capstone Project' (Bayes Business School); 'M&A'(ESADE).
- **Full Scholarships**: NUS Exchange (Jan - Jun 2025), ESADE Exchange (Aug - Dec 2024);
- **Fully-funded scholarship for Agentic AI Bootcamp** by the Beijing University of Posts & Telecommunications (Jun - Jul 2026).
- **Extracurricular Coursework**: Stanford CS229 (Machine Learning), CS230 (Deep Learning), EE178 (Probabilistic Systems); MIT RES.6-012 (Probability); Imperial Mathematics for ML; IBM Applied Data Science Specialization; Statistical Rethinking 2026 (R. McElreath)

EDUCATION

National University of Singapore (NUS), Singapore Jul 2026 - 2027 (Expected)
MSc in NUS School of Computing (Business Analytics), Specialised in Statistics

Bayes Business School (formerly City, University of London), UK Sep 2022 - Jun 2026
BSc International Business (Hons), Specialised in AI and Quantitative Methods; First-Class Honours (Highest Distinction)

UCL Economics & Mathematics Foundation Programme, UK Sep 2021 - Jun 2022
Final grade A · Mathematics 87%, Highest Grade in the stream*

RELEVANT PROJECTS

A/B-Test Experimentation Guardrail: SRM Detection & Causal Inference (Independent Project) [Project Page] [GitHub] 2026

- Built an A/B-test statistical guardrail with an optional Anthropic Claude tool-use agent mode: a bounded multi-turn loop that chooses which of 3 schema-defined tools to call next, with the LLM held off the numerical path so every figure comes from scipy and scikit-learn.
- Designed the default pipeline to route deterministically and call the LLM once, at the end, narrating a code-decided verdict as a plain-English summary; one model call per audit, the statistical core covered by a 35-case pytest suite in CI (Python 3.10–3.12).

J.P. Morgan Quantitative Research: Credit Risk & Inventory Optimisation (Applied Project) [Project Page] [GitHub] 2025

- Architected the analytics codebase as an importable Python package with single-responsibility modules (data loading, threshold evaluation, calibration, RL environment), backed by an 82-test suite and a multi-version CI matrix.
- Built a probability-of-default model under an asymmetric cost matrix, lifting held-out profit by $363K over the default 0.5 threshold, and a tabular Q-learning agent for a finite-horizon inventory-optimisation problem.

Olist Brazilian Marketplace: Causal Inference & Bayesian Modelling (Independent Project) [Project Page] [GitHub] 2026

- Architected the analysis as a clean, importable Python package (ETL, feature builders, causal DAG, separate model factories per likelihood) with a DuckDB feature stack that rebuilds end-to-end in roughly five seconds.
- Validated the models with held-out cross-validation (PSIS-LOO), posterior-predictive checks, and prior-sensitivity sweeps. The headline result shifted just 0.0004 logit across three hyperprior families.

SKILLS

Programming & Databases: Python (NumPy, Pandas, scikit-learn, TensorFlow, PyTorch, statsmodels, SciPy, XGBoost, PyMC), SQL, DuckDB, R, C++.
Machine Learning & AI: Supervised & Unsupervised Learning, Gradient Boosting, Neural Networks (CNNs, RNNs, LSTMs, Transformers, Two-Tower), Reinforcement Learning (Q-learning, PPO, Actor-Critic), Recommendation Systems & Embeddings, LLMs, Prompt Engineering, Agentic AI (LangChain, LlamaIndex, tool-use).
Statistical Methods: Causal Inference, A/B Testing & Experimentation, Hypothesis Testing, Survival Analysis (Cox PH, Random Survival Forest, Kaplan-Meier), Bayesian Statistics, Time-Series Analysis, Probability Calibration, Randomised Controlled Trials.
Visualisation, Cloud & MLOps: Tableau, Power BI, matplotlib, seaborn, Plotly; Google Cloud Platform, AWS, Docker; MLOps (CI/CD, GitHub Actions, model deployment, feature pipelines); Git, Jupyter, LaTeX.
Spoken Languages: English (Fluent), Russian (Native), Ukrainian (Native), Belarusian (Fluent), Spanish (Professional Working).